

Crime in Philadelphia: Bayesian Clustering with Particle Optimization

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Scientific Problem

- Crime in large cities is difficult due to spatial correlation between observations, small-scaled clustering (e.g. zoning), and temporal trends.
- Units are also difficult to deal with. So we care more about crime per capita or crime per unit of area?
- We know how to analyze spatio-temporal data with geospatial data. But what about spatio-temporal with areal data? Clustering? What groups of neighborhoods have similar crime?

Importance

- Important for urban planning, law enforcement, and finding out where crime is actually falling or rising.
- Also... I think people enjoy not being shot. (ADD CITATION)

Literature

- Essentially, the modeling of crime comes down to modeling discontinuity for spatial data.
- Bayesian hierarchical models like CAR, Gaussian Processes, and geographically weighted regression are commonly used for crime data. But their responses are too smooth when there is clustering.
- Other approaches include modeling the discontinuous areas separately or using a smooth model with jumps included at the discontinuity. Too many parameters.

Literature

- Idea: Let $y(s_i) = \alpha(s_i) + \beta(s_i)t$ and cluster the $\alpha(s_i)$ and $\beta(s_i)$ over space.
- Knorr-Held and Raßer (2000)[5], Denison and Holmes (2001)[2], Gangnon and Clayton (2000) [3] and Green and Richardson (2002) [4]. Did this procedure with Poisson regression.
- Teixeira, Assunção [9](2019) and Li and Sang (2019) [6] did spatial clustering by using Bayesian tree methods to model the adjacency structure of the neighborhoods.

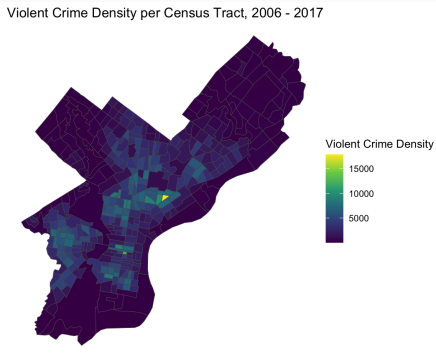
Novelty

Creates a spatial clustering method that

- 1 Allows the spatial distribution of baseline crime to differ from time trends.
- 2 Allows the parameters to vary both within and between clusters.
- 3 Extends Veronika Ročková's method that finds multiple partitions of space with high posterior probability.

The Data

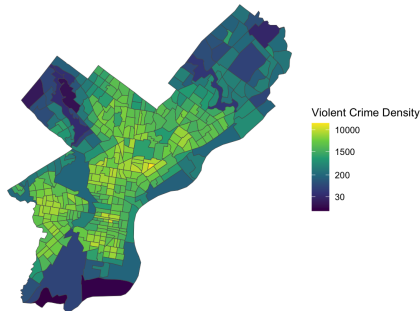
- Outcome is **violent crime density**:
 - Defined as number of violent crimes divided by the land area
 - Violent crimes include homicides, rapes, robberies, and aggravated assault.



Transformation of the Data

- Because the crime densities are heavily skewed, a transformation is necessary for modeling.
- The *inverse hyperbolic sine transformation* is utilized:
 - it behaves similarly to the logarithmic transformation,
 - however, it is well-defined at 0.

Violent Crime Density per Census Tract, 2006 - 2017



Methodology Overview

- The authors propose a "CAR-within-Clusters" model where two latent spatial partitions of the neighborhoods are introduced:
 - One for the baseline levels of crime, and
 - One for the time trends.
- Separate CAR priors are specified on the neighborhood-specific parameters within each cluster of each partition.
- Allows for spatial continuity with the clusters (via the CAR model) but not between clusters (allows for discontinuities)

CAR-within-Clusters Model

- Let the latent partitions of the neighborhoods be $\gamma^{(\alpha)}$ and $\gamma^{(\beta)}$ where $\gamma^{(\cdot)} = \{S_1^{(\cdot)}, \dots, S_{K^{(\cdot)}}^{(\cdot)}\}$ and the sets $S_k^{(\cdot)}$ are the clusters.
- Denote $\gamma := (\gamma^{(\alpha)}, \gamma^{(\beta)})$ as the pair of latent partitions.
- For α (and similarly for β):
 - $\alpha_k | \bar{a}_k, \sigma^2, \gamma^{(\alpha)} \sim \text{CAR}(\bar{a}_k, a_1 \sigma^2, W_k^{(\alpha)})$ for $k = 1, \dots, K_\alpha$
 - $\bar{a}_k \stackrel{\text{iid}}{\sim} N(0, a_2 \sigma^2)$
 - $\gamma^{(\alpha)} \sim \mathcal{T}\text{-}\mathcal{EP}$: truncated Ewens-Pitman prior.

Variational Approximation [8]

- Due to large number of all possible pairs of spatial partitions, computationally infeasible to compute $\pi(\gamma|y)$
- A variational approximation is used to estimate the L pairs of partitions with largest posterior mass, $\Gamma_L = \{\gamma^{(1)}, \dots, \gamma^{(L)}\}$
 - Allows for approximation of expectations of the form $E[g(\alpha, \beta)|y, \gamma]$

Variational Approximation [8]

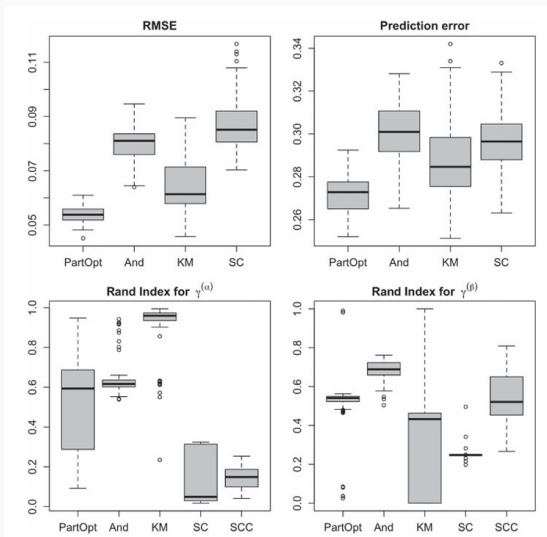
The variational approximation works by:

- Taking $\mathcal{Q}_L$ to be collection of all discrete distributions $q(\cdot|\Gamma, \mathbf{w})$ that place probability w_l on particle γ_l
- For $\lambda > 0$, define the tempered marginal posterior Π_λ with density $\pi_\lambda(\gamma) \propto \pi(\gamma|\mathbf{y})^{\frac{1}{\lambda}}$
- A proposition proven in the supplementary material affirms that the L particles with largest tempered posterior mass $\forall \lambda$ will be those with largest posterior mass, Γ_L

Simulation Study

- The proposed method was investigated across situations ranging from stark to more ambiguous clusters
- Comparisons with several competitors made with respect to RMSE, prediction error, and the adjusted Rand indices between the estimated partitions and the truth
 - Rand index (CITE) between two partitions measures proportion of pairs of items that are clustered together across both partitions

Simulation Study



Case Study

- Transformed density of violent crimes, $y_{i,t}$, is given a Gaussian distribution with mean $\alpha_i + \beta_i x_t$ and variance σ^2
 - Neighborhood i , time t
- Goal: identify two partitions of neighborhoods based on similar mean crime levels, α_i , and time trends, β_i
- Implemented above methodology with $L = 20$ particles

Results

- The top particle returned found many clusters across mean crime density, recovering elements of geography and built environment
- Top particle contains just one cluster of estimated time trends

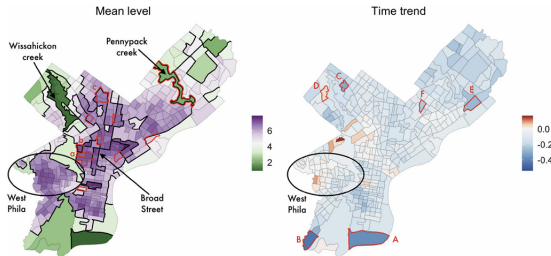


Figure 5. Top particle identified by PartOpt. The thick black lines delineate the borders between the clusters, and the color represents the posterior mean of α and β given the top particle. Regions highlighted in red have different cluster assignment in the particle set.

Results

Table 1. Out-of-sample RMSE for predicting the transformed crime density in 2018 based on models fit to data from 2006–2017.

MAP	PartOpt	MLE	And	KM	SC	SCC
0.2300	0.2299	0.2349	0.2395	0.2341	0.2401	5.970

- Investigated predictive performance via RMSE for predicting crime density in each neighborhood in 2018
- Predictions based on model fit to 2006-2017 data
- Comparisons made to competing methods as well as predictions based solely on MLE of α and β

References

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Truncated Ewens-Pitman Prior

- The truncated Ewens-Pitman prior is utilized within the hierarchical modeling for Bayesian clustering.
- It is characterized by favoring a small number of large clusters and asymptotically induces an average of $\eta \log(N)$ clusters.
- To recover partitions with connected clusters, we truncate this distribution to the set of spatial partitions $\mathcal{SP}$.
- The pmf of this prior is given by

$$\pi(\gamma) \propto \eta^K \prod_{k=1}^K (n_k - 1)! \times \mathbf{1}(\gamma \in \mathcal{SP})$$

and we denote it by $\mathcal{T}\text{-}\mathcal{EP}$.